Assignment-2

(Spring' 2026)

DMP, NSU

MAT 116, Full Marks: 10, Pre-Calculus

Date of Submission: 30 April, 2026

(Answer the following questions sequentially)

1. (a) Use transformations to graph the function:
$f(x) = 2(x-2)^4 + 1$

   (b) Given that $f(x) = 2(x+3)(x^2+4)^3$
Determine the behavior of the graph near each x-intercept (zero) and determine the end behavior that the graph of *f* resembles for large values of $|x|$.

2. Find the vertical,horizontal,and oblique asymptotes,if any,of each rational function:
$$f(x) = \frac{x^3 - 1}{x^2 - 5x - 14}$$

3. Analyze the graph of the function:
$$f(x) = \frac{x^2}{x^2 + x - 6}$$

4. Solve the inequality algebraically:
$$\frac{5}{x-3} \le \frac{3}{x+1}$$

5. Use the Rational Zeros Theorem to find all the real zeros of the polynomial function:
$f(x) = 2x^4 + x^3 - 7x^2 - 3x + 3$

6. (a) Solve: $\log_3(x+1) + \log_3(x+4) = 2$

   (b) Solve: $25^x - 8.5^x = -16$

(The End)